Square-Root Circuit Bounds for Iterated Functions

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Preliminary draft

Abstract

Let $F : \{0,1\}^n \to \{0,1\}^n$ have a Boolean circuit of size s, and put $A = s + n^3$ and $B = A + \sqrt{n^3 2^n A}$. Every fixed iterate F^T has a bounded-fan-in circuit of size $O(B)$, independently of the magnitude of $T \geq 0$, without assuming injectivity. The construction uses sparse checkpoints and sorting-based lookup, with every access and arithmetic gate charged. It is nonuniform and gives no efficient preprocessing algorithm from a transition circuit.

For even n, an explicit linear-size permutation has almost all powers of complexity $\Omega(2^{n/2}/n)$ under a uniform exponent modulo its order. This matches the upper bound's exponential scale, but identifies no explicit hard power; hard exponents necessarily have exponentially many binary digits. The upper-bound construction also gives $O(B)$-gate cycle labels, transient depth, pair-input reachability, exact first hitting times, and complete visit descriptors, together with input-time and shared batched versions.

Structured transitions admit stronger bounds. A permutation differing from a counter at r states has $O(n(r+1))$-gate signed fixed powers and navigation, and $O(n(r+m+1))$-gate iteration for an m-bit input time. Its orbit chart can be constructed in polynomial time from an explicit edit list. An autonomous q-bit controller with a p-bit affine payload gives $O(2^q(q+p^2))$-gate fixed powers. Lean checks the circuit bounds; the edit-list compiler remains outside that boundary. The checkpoint and block methods have established antecedents, and historical priority of these circuit formulations remains unresolved.

Keywords. Boolean circuit complexity; function iteration; functional graphs; nonuniform computation; sorting networks; permutation inversion; one-way functions.

1 Introduction

A simple transition rule can describe a computation of enormous duration. How complicated can its final-state map be as a Boolean function of the initial state? We study this question for a fixed map $F : \{0,1\}^n \to \{0,1\}^n$. The computational resource is the total number of Boolean gates, rather than the number of evaluations of F, the number of memory probes, or the time of a random-access algorithm.

Let $N = 2^n$ and suppose that F has a circuit of size s. Directly unrolling T applications gives a circuit of size at most Ts for F^T. Alternatively, hardwiring a complete truth table and testing the input against all N states gives an $O(nN)$-gate circuit, independently of T. The latter bound ignores the possibility that F itself is much simpler than an arbitrary map on its state space. Our main result exploits that possibility.

Theorem 1.1 (Fixed-power tradeoff)**.** *Let $n \geq 1$, let $F : \{0,1\}^n \to \{0,1\}^n$ satisfy $\mathrm{C}(F) \leq s$, and put $N = 2^n$ and $A = s + n^3$. For every integer $T \geq 0$ and every integer $1 \leq L \leq N$,*

$$\mathrm{C}(F^T) = O\Big(Ls + n^3\left(L + \frac{N}{L}\right)\Big). \tag{1.1}$$

In particular,

$$\mathrm{C}(F^T) = O\Big(A + \sqrt{n^3 NA}\Big). \tag{1.2}$$

The implicit constants are absolute. The circuit may depend on F and T, but the size bound does not depend on the bit length of T.

Here and throughout, circuits use the full fan-in-two Boolean basis and have unrestricted fan-out. The results also hold over the basis $\{\mathsf{AND}, \mathsf{OR}, \mathsf{NOT}\}$ with a constant-factor change. Circuit size refers to the whole multi-output circuit, not to a separate size bound for each output bit.

Corollary 1.2. *If $\mathrm{C}(F_n) = n^{O(1)}$, then*

$$\sup_{T \geq 0} \mathrm{C}(F_n^T) \leq 2^{n/2 + O(\log n)}. \tag{1.3}$$

The constant hidden in $O(\log n)$ may depend on the polynomial bound for $\mathrm{C}(F_n)$, but not on T.

For each fixed F and T, the circuit works for every initial state.

1.1 Proof overview

The functional graph of F consists of directed cycles with directed trees feeding into them. For a parameter L, we select $O(N/L)$ checkpoints so that every trajectory either encounters one within $2L$ steps or repeats a state within that many steps. A repeated state reveals a period and permits direct recovery of the answer from the initial trajectory.

Set $K = 2L$ and assume $T \geq K$. At each checkpoint h, store $z_h = F^{T-K}(h)$. If the initial trajectory encounters h at time $d \leq K$, then

$$F^{K-d}(z_h) = F^{K-d}(F^{T-K}(F^d(x))) = F^T(x). \tag{1.4}$$

Thus only a short *forward* correction is needed after the lookup. There is no inverse operation, so the argument applies to noninjective maps.

A separate lookup for each of $\Theta(L)$ visited states against $\Theta(N/L)$ checkpoints would cost $\Theta(N)$ comparisons. Instead, we generate the full initial trajectory before looking anything up and resolve all its queries together. Sorting the queries and dictionary records reduces this cost to $\mathrm{poly}(n)(L + N/L)$ Boolean gates. Combining the initial and final short simulations with this batched lookup gives Equation (1.1).

A worked example. Consider eight states, encoded by three bits, with transitions

$$0 \to 1 \to 2 \to 3 \to 4 \to 5 \to 2, \qquad 6 \to 7 \to 6.$$

Take $L = 3$, $K = 6$, and $H = \{2, 5\}$, the endpoints of the disjoint three-vertex paths $(0, 1, 2)$ and $(3, 4, 5)$. Fix $T = 101$. Starting at $x = 1$, the first checkpoint is $h = 2$, reached at $d = 1$. Its stored payload is $z_2 = F^{95}(2) = 5$, so the forward correction gives

$$F^{K-d}(z_2) = F^5(5) = 2 = F^{101}(1).$$

Starting instead at $x = 6$, the prefix $6, 7, 6$ repeats without meeting a checkpoint. The repeated indices $i = 0$, $j = 2$ give period $p = 2$. The answer is prefix entry $i + ((101 - i) \bmod p) = 1$, namely 7. Thus the two-cycle needs no dictionary entry.

1.2 Further results and sharpness

The exponent $1/2$ is sharp for arbitrary fixed powers. In Section 4, an explicit linear-size permutation has almost all powers of complexity $\Omega(2^{n/2}/n)$, when the exponent is uniform modulo its order. The hard powers are selected by counting, require exponentially long binary exponents, and are not explicitly identified. The same permutation has a polynomial-size iterator for polynomial-width input times.

Write $B = A + \sqrt{n^3NA}$. The extensions in Sections 5.1 and 5.2 handle input exponents and shared evaluation. The graph-query circuits in Section 6 compute cycle labels, transient depth, reachability, first hitting times, visit descriptors, and synchronous coalescence in $O(B)$ gates. Pair queries take two n-bit states; their exponential scale remains $2^{n/2}$ for polynomial-size transitions. Recognizable targets include eventual membership and recurring or permanent acceptance. For a predicate of size a, put $A_P = s + a + n^3$ and $B_P = A_P + \sqrt{n^3NA_P}$.

Section 7 gives stronger bounds for autonomous controllers and sparse counter edits. For the latter, an explicit edit list permits polynomial-time chart construction; transport to other coordinates charges their encoding and decoding circuits. Choosing $T = \operatorname{ord}(P) - 1$ also recovers the earlier square-root permutation-inversion bound, compatible with polynomial-size one-wayness; see Section 3.6.

Query	Input parameters	Circuit size
General transitions on n-bit states		
Fixed iterate	x; fixed $T \geq 0$	$O(B)$
Input-time iterate	x, t; $t \in \{0,1\}^m$	$O(\min\{mB, nB + nm\})$
Graph navigation	x or x, y	$O(B)$
Recognizable targets	x; fixed P of size a	$O(B_P)$
Shared fixed powers	k states; common fixed T	$O(kA + \sqrt{kn^3NA})$
Autonomous q-bit controller and p-bit affine payload, $q + p = n$		
Fixed iterate	u, v; fixed $T \geq 0$	$O(2^q(q + p^2))$
Counter permutations with r edited successors		
Signed fixed iterate; labels and navigation	x or x, y; fixed $T \in \mathbb{Z}$	$O(n(r + 1))$
Input-time iterate	x, t; $t \in \{0,1\}^m$	$O(n(r + m + 1))$

Table 1: Circuit bounds for fixed transitions, with $m \geq 1$ and $1 \leq k \leq N$. An m-bit input time multiplies the affine-controller bound by m. Visit descriptors specify all visits; additional deadline-processing circuits are not included. All listed bounds are formalized; Section 8 records the checked boundary.

1.3 Related work and status of the formulation

Preprocessing trajectories to exchange storage for later computation is classical. Hellman's cryptanalytic time–memory tradeoff is an early example of this approach [3]. More directly, Munro, Raman, Raman, and Rao [4, Section 3.1] give a shortcut representation for permutation inversion: with a parameter L, their black-box construction uses $O(L)$ evaluations and $O((N/L)(\log N + \log L))$ auxiliary bits. They also give succinct representations of general functions supporting positive powers with a word-size exponent [4, Section 5]. Those results concern representation space, RAM time, and black-box access, rather than total Boolean gate count.

A closer circuit antecedent is Hirahara, Ilango, and Williams [5, Section 3.2]: they obtain square-root-size permutation-inversion circuits using checkpoints and batched lookup. Their lookup theorem builds on a lemma of Paul. This draft generalizes that permutation-inversion construction to arbitrary fixed forward powers of possibly noninjective maps. It does not subsume their general-function inversion and compression theorems.

Eppstein [6] studies uniform iteration of polynomial-time bijections, with the iteration count supplied in binary, and characterizes the resulting class of functional problems as $\mathsf{FP}^{\mathsf{PSPACE}}$. Our resource question differs: the map is fixed, the circuits are nonuniform, and the size bound remains exponential in the state length. Our result supplies no polynomial-time algorithm for that iteration problem.

The present construction uses the same checkpoint and batched-lookup principles. Batcher's sorting networks [1] implement the lookup; the asymptotically smaller networks of Ajtai, Komlós, and Szemerédi [2] save a factor of n (Section 5.3), while the shifted payload in Equation (1.4) and the checkpoint-or-repetition lemma handle general powers and functional graphs. No RAM memory probe is treated as a unit-cost circuit operation. Checkpointing, sorting-based lookup, and square-root-size permutation inversion are not claimed as new. Historical priority for the full fixed-power theorem and the shared bounds in this draft remains unresolved.

2 Model and circuit primitives

2.1 Conventions

All logarithms are to base two. The notation F^t denotes t-fold composition, with F^0 equal to the identity. The quantity $\mathrm{C}(G)$ is the minimum number of internal gates in an acyclic Boolean circuit computing all output bits of G. Each internal gate computes a Boolean function of at most two bits. Inputs, wires, fan-out, and output designations are not charged as gates. The two constant bits are available; charging a constant number of gates for them would not change any bound.

The circuit for an iterate is allowed to depend on the entire function F and on its fixed exponent. Preprocessing time and the time required to find the circuit are not bounded. Hardwired data do not provide free addressing: every comparison, multiplexer, and routing operation used to access them is realized with Boolean gates and included in our estimates.

All state words have n bits. Because $L \le 2^n$, indices up to $2L$ fit in $n+2$ bits. In the batched construction $r \le 2^n$, so query identifiers and record positions also have $O(n)$ bits. We retain explicit polynomial factors in n for a concrete implementation; these factors are not claimed to be optimal.

2.2 Arithmetic and selection

Comparison, addition, subtraction, and conditional selection of w-bit words have $O(w)$-gate circuits. Binary long division gives an $O(w^2)$-gate circuit for the remainder of a w-bit integer modulo a positive w-bit integer: process the dividend bits from most to least significant, and at each stage conditionally subtract the modulus from the current shifted remainder. There are $O(w)$ stages and $O(w)$ gates per stage.

A circuit can select one of q words of width w using $O(qw)$ gates by a binary tree of word multiplexers, padding to a power of two when necessary. It can also select one flagged record from an ordered list of q records in $O(qw)$ gates. For example, a left-to-right scan maintains a bit indicating whether a flagged record has already been found and retains the first one. This scan is simply an acyclic chain of gates; no depth bound is needed here.

Lemma 2.1 (Batched fixed-dictionary lookup)**.** *Suppose a fixed dictionary consists of M distinct w-bit keys, each with a w-bit payload. Given $Q \geq 1$ query keys, a Boolean circuit can output a match bit and the corresponding payload for every query, in the original query order, using*

$$O\Big((M+Q)(w+\log(M+Q+2))\log^2(M+Q+2)\Big) \tag{2.1}$$

gates. The payload of an unmatched query can be set to zero. In particular, if $w = O(n)$ and $\log(M+Q+2) = O(n)$, the cost is $O(n^3(M+Q))$.

Proof. Let $R = M + Q$. Form one record for each dictionary entry and one for each query. A record contains a key, a dictionary/query tag, a payload field, and an original-position field. The record width is $O(w + \log(R+2))$.

Sort the records lexicographically by key, then by the tag with dictionary records first, and then by original position. A sorting network with $O(R\log^2(R+2))$ comparators exists [1]. A comparator and the resulting conditional swap of the entire records cost $O(w + \log(R+2))$ Boolean gates. Padding uses a separate dummy-record bit so that no actual key is reserved as a sentinel.

In the sorted list, a dictionary record precedes every query with its key. Scan the list, resetting the match bit when the key changes and propagating the payload from a dictionary record through the equal-key run that follows it. If an equal-key run has no dictionary record, every query in that run receives match bit zero. Dictionary keys are distinct, so no run contains two competing payloads. This scan uses $O(R(w + \log(R+2)))$ gates.

Finally, sort records by a tag that places queries first and then by the query's original position. The first Q records give the requested outputs in order. A second sorting network has the same asymptotic cost as the first. Every operation described is a fixed acyclic circuit. □

Lemma 2.2 (Detecting a repeated state)**.** *Given Q words of w bits, a circuit can return a flag indicating whether two words are equal and, when the flag is one, indices $i < j$ of two equal words. Its size is*

$$O\Big(Q(w+\log(Q+2))\log^2(Q+2)\Big).$$

In particular, for $Q \leq 2^{n+O(1)}$ and $w = O(n)$, the size is $O(n^3Q)$.

Proof. Attach each word's original index and sort by word and then by index. A duplicate exists exactly when two adjacent sorted records have the same word. Compare adjacent words and select one flagged pair. Sorting by index within an equal-word run guarantees that the returned indices satisfy $i < j$. The remaining comparisons and selection use at most $O(Q(w+\log(Q+2)))$ gates. □

2.3 Sparse checkpoints in functional graphs

The functional graph of a map F has one vertex for each state and the edge $v \to F(v)$ at every vertex. Each weakly connected component has a unique directed cycle, with directed trees feeding into it. For a noncycle vertex, its *depth* is the number of steps needed to reach the cycle.

Lemma 2.3 (Checkpoint or repetition)**.** *Let F be a function on a set of N states, and let $1 \leq L \leq N$. There is a set H of at most $\lfloor N/L \rfloor$ states such that, for every x, either two of $x, F(x), \ldots, F^{L-1}(x)$ are equal or $F^d(x) \in H$ for some $d < 2L$. More generally, the same conclusion holds for starting states in any finite forward-invariant set S, with $|H|L \leq |S|$ and $H \subseteq S$. The ambient state type need not be finite.*

Proof. Choose a maximal family of pairwise vertex-disjoint simple directed paths, each consisting of L vertices in S. Put the last vertex of each path into H. Disjointness gives $|H|L \le |S|$. If the first L states of an orbit are distinct, they form such a path in S. By maximality this path meets a chosen path: say $F^i(x) = F^j(a)$, where $i, j < L$ and the chosen path starts at a. Its marked endpoint is reached from x after $i + (L - 1 - j) < 2L$ steps. If the prefix is not simple, it already contains a repetition. Taking S to be the entire state space proves the first claim. The Lean proof implements the same packing argument by induction on $|S|$, removing one simple path at a time. □

Remark 2.4. It is essential that the lemma allows a short repetition instead of a checkpoint. A function can have N fixed points, in which case a set meeting every eventual cycle must contain all N states. Such cycles need no stored shortcut: their behavior is already visible locally.

3 Circuits for fixed powers

We now prove Theorem 1.1. The construction is uniform as a description of a circuit from the chosen constants, but no efficient procedure for computing those constants is asserted.

3.1 Short trajectories and shifted payloads

Fix L and let $K = 2L$. If $T < K$, use T copies of a circuit for F; this costs at most $2Ls$ gates. Hence assume $T \ge K$. Choose H as in Lemma 2.3. For every $h \in H$, hardwire the n-bit payload

$$z_h = F^{T-K}(h). \tag{3.1}$$

The dictionary is fixed independently of the input x and has at most N/L entries.

On input x, first compute

$$x_0 = x, \quad x_1 = F(x), \quad \ldots, \quad x_K = F^K(x). \tag{3.2}$$

This requires Ks gates. Apply Lemma 2.1 to all $K + 1$ states in Equation (3.2), using the dictionary $h \mapsto z_h$. Select one matching record, if any, and retain its original trajectory index d. This gives a match flag and a pair (d, z_h) with $x_d = h$ whenever the flag is one.

For the matched case, compute a second short trajectory

$$z_h, F(z_h), \ldots, F^K(z_h)$$

and select its entry at position $K - d$. Its value is

$$\begin{aligned} F^{K-d}(z_h) &= F^{K-d}\big(F^{T-K}(h)\big) \\ &= F^{T-d}(h) \\ &= F^{T-d}\big(F^d(x)\big) = F^T(x). \end{aligned} \tag{3.3}$$

All exponents used here are nonnegative. This second trajectory costs another Ks gates; selection costs $O(nK)$ gates.

3.2 The repeated-state case and very large exponents

Use Lemma 2.2 on Equation (3.2). If no checkpoint was found, Lemma 2.3 guarantees indices

$$0 \le i < j \le K, \qquad x_i = x_j.$$

Put $p = j - i$. Determinism implies $x_{u+p} = x_u$ for all $u \ge i$. The number p need not be the least period. Since $T \ge K \ge i$,

$$F^T(x) = x_{\,i+((T-i) \bmod p)}. \tag{3.4}$$

The selected index is at most $i + p - 1 = j - 1$, so the required state already occurs in the initial trajectory.

To implement Equation (3.4) without paying for the bit length of T, hardwire the table

$$R_T(p) = T \bmod p, \qquad 1 \le p \le K. \tag{3.5}$$

There are K entries, each of $O(n)$ bits. A multiplexer tree retrieves $R_T(p)$ in $O(nK)$ gates. Compute $i \bmod p$ by binary long division and then compute

$$(T - i) \bmod p = \big(R_T(p) - (i \bmod p)\big) \bmod p$$

using a comparison, subtraction, and a conditional addition of p. These operations take $O(n^2)$ gates. A further $O(nK)$ gates select the state prescribed in Equation (3.4).

A circuit evaluates both branches, so all intermediate operations must also be defined when their output will be discarded. If a repeated pair is absent, supply the dummy values $i = 0$ and $p = 1$ to the arithmetic branch. If a checkpoint is absent, supply a dummy state and $d = 0$ to the second trajectory. The final output selects the checkpoint branch when its match flag is one, and otherwise selects the repetition branch. The checkpoint lemma ensures that the selected branch is always valid. In particular, no division by zero occurs on any input.

3.3 Size bound

The two simulations together use at most $2Ks = 4Ls$ gates. The dictionary and query list contain

$$|H| + (K + 1) = O\!\left(\frac{N}{L} + L\right)$$

records of $O(n)$ bits, and the logarithm of this number is $O(n)$. Thus Lemma 2.1 costs $O(n^3(N/L+L))$ gates. Repeated-state detection costs $O(n^3L)$ gates, and all remaining lookup, arithmetic, and selection operations cost $O(nL + n^2)$ gates. Since $n, L \ge 1$, the latter cost is absorbed by $O(n^3L)$. This proves Equation (1.1).

Write the tradeoff as $O(AL + n^3N/L)$, where $A = s + n^3$, and choose

$$L = \max\left\{1, \left\lceil \sqrt{n^3N/A} \right\rceil \right\}. \tag{3.6}$$

Since $A \ge n^3$, this choice satisfies $L \le \left\lceil \sqrt{N} \right\rceil \le N$. If $A \le n^3N$, then $\sqrt{n^3N/A} \le L \le 2\sqrt{n^3N/A}$. Hence

$$AL = O(\sqrt{n^3NA}), \qquad n^3N/L = O(\sqrt{n^3NA}).$$

If $A > n^3N$, the choice is $L = 1$ and the cost is $O(A)$. This establishes Equation (1.2) and completes the proof of Theorem 1.1. □

Remark 3.1 (Where the exponent is stored). For $T \geq K$, the information about T used by the circuit is confined to the checkpoint payloads in Equation (3.1) and the residue table in Equation (3.5). Their total length is $O(n(N/L + L))$ bits. The construction does not place the binary expansion of T on a long hidden input tape, and it does not perform uncharged arithmetic on an arbitrarily large integer. Computing the hardwired information is a preprocessing task outside the asserted circuit bound.

3.4 Charging only long trajectories

Let $\delta(v)$ be the distance from v to its eventual cycle, and let $\ell(v)$ be that cycle's length. For $1 \leq L \leq N$, put

$$Q_L = |\{v : \delta(v) \geq L\}| + |\{v : \delta(v) = 0,\ \ell(v) \geq L\}|.$$

The two sets are disjoint. Their union contains the deep transient vertices and all vertices of cycles with length at least L.

Theorem 3.2 (Long-trajectory tradeoff)**.** *For every fixed $T \geq 0$ and $1 \leq L \leq N$,*

$$\mathrm{C}(F^T) = O\Big(Ls + n^3\left(L + \frac{Q_L}{L}\right)\Big).$$

Proof. Apply the path-packing argument of Lemma 2.3 only to paths whose L vertices all lie in the union above. Mark their endpoints. The resulting set H satisfies $|H|L \leq Q_L$. Every L-state prefix inside this union either repeats or meets a packed path, and hence reaches its marked endpoint within $2L$ steps. This packing argument does not require the union to be forward invariant.

If $\delta(x) \geq 2L$, the first L states are all deep, so that argument applies immediately. Otherwise the trajectory reaches its cycle before $2L$ steps. A short cycle then repeats before $3L$ steps. On a long cycle all vertices lie in the union, so the packing argument gives a checkpoint or repetition within another $2L$ steps. Thus every trajectory hits H or repeats within a prefix of $4L$ steps.

Use this prefix in the fixed-power construction, with payload $F^{T-4L}(h)$ when $T \geq 4L$ and direct unrolling otherwise. The dictionary has at most Q_L/L entries and the prefix has $O(L)$ states, giving the stated cost. □

Short transient paths leading to short cycles incur no dictionary cost. The bound includes $Q_L = 0$. Since Q_L itself depends on L, one must retain that dependence when choosing the tradeoff parameter.

3.5 Invariant sets and image-sensitive bounds

Theorem 3.3 (Invariant-set tradeoff)**.** *Let $S \subseteq \{0,1\}^n$ be forward invariant under F, with $|S| = M$. For every fixed $T \geq 0$ and $1 \leq L \leq N$, there is a circuit agreeing with F^T on S of size*

$$O\Big(Ls + n^3\left(L + \frac{M}{L}\right)\Big).$$

The circuit is total on $\{0,1\}^n$; correctness outside S is not asserted.

Proof. Use the invariant-set version of Lemma 2.3, and otherwise apply the fixed-power construction unchanged. A state still occupies n bits, but the dictionary contains only $O(M/L)$ records. □

Corollary 3.4 (Image sensitivity)**.** *Put* $M_d = |\operatorname{im}(F^d)|$*. For* $T \geq d$ *and* $1 \leq L \leq M_d$*,*

$$\mathrm{C}(F^T) \leq \mathrm{C}(F^d) + O\Big(Ls + n^3\left(L + \frac{M_d}{L}\right)\Big).$$

Consequently this is at most $\mathrm{C}(F^d) + O(A + \sqrt{n^3 M_d A})$*.*

Proof. The image of F^d is forward invariant, since $F(F^d(x)) = F^d(F(x))$. Compute $F^d(x)$ and apply Theorem 3.3 on its image for $T - d$ more steps. Optimize L as in Equation (3.6), replacing N by M_d. □

3.6 Permutation inversion and one-way functions

Corollary 3.5 (Exact permutation inversion)**.** *Let* $P : \{0,1\}^n \to \{0,1\}^n$ *be a permutation with* $\mathrm{C}(P) \leq s$*, and put* $N = 2^n$ *and* $A = s + n^3$*. For every* $1 \leq L \leq N$*,*

$$\mathrm{C}(P^{-1}) = O\Big(Ls + n^3\left(L + \frac{N}{L}\right)\Big).$$

In particular,

$$\mathrm{C}(P^{-1}) = O\Big(A + \sqrt{n^3 N A}\Big).$$

If $s = n^{O(1)}$*, an inverter of size* $2^{n/2+O(\log n)}$ *is correct on every input.*

Proof. Let $D = \operatorname{ord}(P)$, the least common multiple of its cycle lengths. Then P^D is the identity, so $P^{-1} = P^{D-1}$. Apply Theorem 1.1. The exponent may have a long binary expansion, but the theorem charges only the resulting tables and their access gates. □

This recovers the permutation-inversion circuit bound already discussed by Hirahara, Ilango, and Williams [5, Section 3.2]. It identifies an earlier special case of the fixed-power theorem.

One-wayness against polynomial-size circuits requires every such inverter to have negligible success on the image of a uniform input [7, Section 7.1.1]. The exponential-size exact inverter above is compatible with that requirement. It does rule out security against all circuits of size $2^{(1/2+\varepsilon)n}$, for any fixed $\varepsilon > 0$ and sufficiently large n; it establishes no security at smaller sizes.

For noninjective maps, a forward power need not find a preimage. For example, the map with values $(0, 0, 1, 0)$ on $\{0, 1, 2, 3\}$ has unique preimage 2 of 1, while the forward orbit of 1 stays in $\{0, 1\}$. Hirahara, Ilango, and Williams give a separate general-function inversion bound of $2^{4n/5}\operatorname{poly}(s + n)$ gates [5, Theorem 3.5]. Encoding exhaustive preimage search as iteration retains both an n-bit target and an n-bit candidate, so the present theorem gives only $2^{n+O(\log n)}$ gates for that construction.

Nor does the counting example in Section 4 establish one-way functions. Its update permutation has a linear-size inverse, obtained by decrementing the active cycle coordinate. The selected hard powers have exponential worst-case forward complexity, so cannot themselves form a polynomial-time one-way function. The counting argument gives no average-case inversion hardness.

4 A matching exponent for arbitrary fixed powers

This section gives an explicit linear-size permutation whose collection of powers is large enough that some member has circuit complexity at least $2^{n/2}/\operatorname{poly}(n)$. The counting argument is elementary and does not require an asymptotic estimate for the maximal order of a permutation.

4.1 A permutation with cycles of all lengths up to $2^{n/2} - 1$

Let $n = 2k$ and $B = 2^k - 1$. Write a state as a pair of k-bit integers (a, b). Define

$$P_n(a,b) = \begin{cases} (a, b+1 \bmod a), & 1 \le a \le B \text{ and } 0 \le b < a, \\ (a,b), & \text{otherwise.} \end{cases} \tag{4.1}$$

For each $1 \le a \le B$, the states $(a,0), (a,1), \ldots, (a,a-1)$ form one cycle of length a; every other state is fixed. Hence P_n is a permutation and

$$\operatorname{ord}(P_n) = D_B := \operatorname{lcm}(1, 2, \ldots, B). \tag{4.2}$$

Despite the notation in Equation (4.1), the update does not require a general division circuit. Compare b with a, increment b, and, when $b < a$, replace b by zero exactly when the increment equals a. Otherwise retain the appropriate increment or original value. Comparison, increment, and selection each have $O(k)$-gate circuits. The first coordinate is passed through unchanged. Thus

$$\mathrm{C}(P_n) = O(n), \tag{4.3}$$

and the family has a direct polynomial-time circuit construction.

Theorem 4.1 (Powers and fixed-integer remainders)**.** *For a fixed $T \ge 0$, define $r_T(a) = T \bmod a$ when $a > 0$ and $r_T(0) = 0$, using k bits for both input and output. Then*

$$\mathrm{C}(r_T) \le \mathrm{C}(P_n^T) \le \mathrm{C}(r_T) + O(n).$$

Proof. Restrict the power circuit to inputs $(a, 0)$ and retain its second coordinate. This computes r_T, including the zero-modulus branch. Conversely, on an active state compute $b + r_T(a)$ using $k+1$ bits. Both summands are below a, so a comparison and at most one subtraction of a give the result modulo a. The spare bit retains the carry. Select the original b on inactive states and copy a to the output. All additional operations have linear-size Boolean circuits. The first inequality uses the paper's free constant-wire convention; the native Lean model charges one shared zero gate for the restriction. □

Thus the arithmetic object behind the lower bound is the remainder of a single fixed integer as a function of its input modulus. The theorem does not identify a particular hard integer.

Lemma 4.2. *For every integer $B \ge 1$,*

$$\operatorname{lcm}(1, 2, \ldots, B) \ge \frac{2^B}{B+1}. \tag{4.4}$$

Proof. We show that every binomial coefficient $\binom{B}{j}$ divides D_B. For a prime p, factorial valuations give

$$\nu_p\binom{B}{j} = \sum_{a \ge 1} \left(\left\lfloor \frac{B}{p^a} \right\rfloor - \left\lfloor \frac{j}{p^a} \right\rfloor - \left\lfloor \frac{B-j}{p^a} \right\rfloor \right).$$

Each summand is zero or one, and it is zero when $p^a > B$. Therefore

$$\nu_p\binom{B}{j} \le \left\lfloor \log_p B \right\rfloor = \nu_p(D_B),$$

where both sides are zero for $p > B$. This proves divisibility. Since $\sum_{j=0}^{B} \binom{B}{j} = 2^B$, some binomial coefficient is at least $2^B/(B+1)$, and D_B is at least that large. □

Lemma 4.3 (Counting multi-output circuits)**.** *The number of functions $\{0,1\}^n \to \{0,1\}^n$ computable with at most q fan-in-two Boolean gates is at most*

$$(q+1)16^q(n+q+2)^{2q+n} = 2^{O((q+n)\log(n+q+2))}. \tag{4.5}$$

Proof. Topologically order the gates. At each gate choose one of at most 16 Boolean operations and two predecessors among the inputs, the two constants, and gates. Allowing all $n+q+2$ possible predecessors at each position only increases the count. Choose the n output wires from these same possibilities. Sum over gate counts from zero through q; each summand is bounded by the expression without its factor $q+1$. Different encodings may compute the same function, which only helps the upper bound. □

Theorem 4.4 (Hard powers of a simple permutation)**.** *For all sufficiently large even n, the permutation in Equation* (4.1) *satisfies*

$$\max_{T\geq 0} \mathrm{C}(P_n^T) = \Omega\left(\frac{2^{n/2}}{n}\right). \tag{4.6}$$

Together with Theorem 1.1, this gives

$$\log_2\left(\max_{T\geq 0} \mathrm{C}(P_n^T)\right) = \frac{n}{2} + O(\log n). \tag{4.7}$$

Proof. There are exactly D_B distinct powers of P_n, with representatives $0 \leq T < D_B$. By Lemma 4.2,

$$\log_2 D_B \geq B - \log_2(B+1).$$

Let $q = \lfloor cB/n \rfloor$ for a sufficiently small positive absolute constant c. Since $B = 2^{n/2}-1$, Lemma 4.3 bounds the logarithm of the number of circuits of size at most q by

$$O((q+n)\log(n+q+2)) = O(cB + n^2).$$

Choose c so that the coefficient of B is less than $1/2$. For all sufficiently large even n, this quantity is smaller than $B - \log_2(B+1)$. Thus there are fewer size-q circuits than distinct powers of P_n. At least one power has complexity greater than q, proving Equation (4.6).

For the upper bound, substitute $\mathrm{C}(P_n) = O(n)$ into Equation (1.2). This gives $\max_T \mathrm{C}(P_n^T) = O(n^3 2^{n/2})$. Combining the upper and lower bounds and taking logarithms proves Equation (4.7). □

Theorem 4.5 (Almost every power is hard)**.** *There is an absolute $c > 0$ such that, for all sufficiently large even n,*

$$\Pr_{T\sim \mathrm{Unif}\{0,\ldots,D_B-1\}}\left[\mathrm{C}(P_n^T) \geq c\frac{2^{n/2}}{n}\right] \geq 1 - 2^{-\Omega(2^{n/2})}.$$

Proof. Set $U = 2^{n/2}$. The preceding counting argument bounds the number of powers with circuits of size at most $\lfloor cU/n \rfloor$ by $2^{\alpha U}$, for some fixed $\alpha < 1$ after decreasing c. On the other hand, $D_B \geq 2^{U-1}/U$. Divide the two counts to bound the exceptional fraction by $U2^{1-(1-\alpha)U} = 2^{-\Omega(U)}$. □

The probability concerns exponent residues modulo the whole group order. It does not concern short exponent descriptions or average-case hardness over inputs to any one power. In particular, easy powers remain among the exceptional residues.

Corollary 4.6. *For all sufficiently large even n, some power of P_n has an output coordinate of circuit complexity $\Omega(2^{n/2}/n^2)$.*

Proof. For the power supplied by Theorem 4.4, join optimal circuits for all n output coordinates. Their total size is at most n times the largest coordinate complexity and is an upper bound on the complexity of the whole map. □

4.2 Why the lower bound does not apply to short input times

The lower bound selects T from $\{0, \ldots, D_B - 1\}$; it places no polynomial bound on the bit length of that exponent. Indeed, Lemma 4.2 shows that $\log D_B \geq 2^{n/2} - O(n)$. The choice of a power therefore has exponentially many available bits. The theorem does not provide a polynomial-time procedure that, given n, identifies a hard power.

For the very same family, the iterator with an m-bit time input has size $O(n(m+1))$. On an active cycle,

$$P_n^t(a, b) = (a, (b + (t \bmod a)) \bmod a). \tag{4.8}$$

An m-stage remainder computation with a k-bit running remainder costs $O(mk)$ gates, and the remaining comparison, addition, and selection cost $O(n)$ gates as in Theorem 4.1. On inactive states the answer is the input itself; use a dummy positive modulus when $a = 0$. Consequently, for $m = \mathrm{poly}(n)$ this iterator has polynomial size.

Corollary 4.7 (Bit length of hard exponents)**.** *Let T_n be exponents satisfying $\mathrm{C}(P_n^{T_n}) = \Omega(2^{n/2}/n)$ for sufficiently large even n. Their binary lengths $m_n = \max\{1, \lceil \log_2(T_n + 1) \rceil\}$ satisfy*

$$m_n = \Omega(2^{n/2}/n^2).$$

Proof. Hardwire the m_n-bit exponent into the iterator just described. This gives $\mathrm{C}(P_n^{T_n}) = O(n(m_n + 1))$. Comparing with the assumed lower bound proves the claim. □

Thus the hard powers necessarily require exponentially long binary exponents; long descriptions are not merely allowed by the counting proof. Theorem 4.4 establishes sharpness for arbitrary fixed powers. It gives no hardness result for the bounded-bit-time problem.

4.3 Explicit hard exponents

By Corollary 4.7, a hard power of P_n has an exponent with $2^{n/2}/\mathrm{poly}(n)$ binary digits. The natural notion of an explicit hard power therefore allows time polynomial in that length. Call a sequence of exponents $(T_n)_{n \text{ even}}$ *explicit* if a deterministic algorithm outputs the binary expansion of T_n from 1^n in time $2^{O(n)}$. Write $\mathsf{E} = \mathsf{DTIME}(2^{O(\ell)})$ for languages of strings of length ℓ, and $\mathsf{SIZE}(s(\ell))$ for languages whose length-ℓ slices have circuits of size $s(\ell)$.

Theorem 4.8 (Explicit hard powers give circuit lower bounds for E)**.** *Let $\varepsilon > 0$, and let (T_n) be explicit exponents with $\mathrm{C}(P_n^{T_n}) \geq 2^{\varepsilon n}$ for infinitely many even n. Then for every $0 < \delta < \varepsilon$ there is a language in E whose length-ℓ slice requires circuits of size $2^{\delta\ell}$ for infinitely many ℓ.*

Proof. For even $n = 2k$ put $j_n = \lceil \log_2 n \rceil$ and $\ell_n = n + j_n$; these lengths are strictly increasing in n. A string of length ℓ_n is read as (a, b, i) with $a, b \in \{0,1\}^k$ and $i \in \{0,1\}^{j_n}$. Let L contain (a, b, i) exactly when $i < n$ and bit i of $P_n^{T_n}(a, b)$ is one, and let L contain no strings of other lengths.

To decide L, recover n from ℓ, compute T_n in time $2^{O(n)}$, and evaluate $P_n^{T_n}(a, b)$ by Equation (4.8), returning (a, b) on inactive states. The remainder $T_n \bmod a$ of an integer with $2^{O(n)}$ digits by a k-bit modulus takes time $2^{O(n)}$ by long division. Hence $L \in \mathsf{E}$.

Conversely, hardwiring $i = 0, \ldots, n-1$ into n copies of a circuit for the slice L_{ℓ_n} computes all output bits of $P_n^{T_n}$. Thus $\mathrm{C}(L_{\ell_n}) \geq \mathrm{C}(P_n^{T_n})/n \geq 2^{\varepsilon n}/n$ for the infinitely many n in the hypothesis. Since $\ell_n \leq n + \log_2 n + 1$, this exceeds $2^{\delta \ell_n}$ for all sufficiently large such n. □

Unconditionally exhibiting an explicit hard power of P_n is therefore at least as hard as proving an exponential circuit lower bound for E, an open problem; the hypothesis that such a lower bound

holds is the hardness assumption of Impagliazzo and Wigderson [14]. This complements Theorem 4.5: almost every residue modulo D_B is hard, but no efficient procedure is known for producing one.

The amount of hardwired information is also forced. Distinct residues of T modulo D_B give distinct powers, and Theorem 4.1 recovers the remainder table r_T, hence T mod D_B, from P_n^T. Any description of P_n^T from which the map can be evaluated, whether a circuit, a table, or a checkpoint dictionary, therefore determines T mod D_B. By Lemma 4.2, for some T such a description has at least $\log_2 D_B \geq 2^{n/2} - O(n)$ bits. The dictionary of Section 3, with $O(n2^{n/2})$ bits at the balanced parameter, is within a polynomial factor of this minimum.

The converse direction. Does an exponential lower bound for E conversely yield an explicit hard power of P_n? The obvious encoding gives a conditional form, which an index-to-prime circuit makes unconditional for lower bounds above $2^{\ell/2}$. Let $q_m(z)$ denote the $(z+1)$st prime, for $z \in \{0,1\}^m$ read as an integer. By the prime number theorem $q_m(z) < 2^{m+\log_2 m+O(1)}$, and the map q_m is computable in time $2^{O(m)}$ by a sieve. Let k_m be the least integer with $2^{k_m} - 1 \geq q_m(2^m - 1)$; then $k_m = m + \log_2 m + O(1)$, and we regard q_m as a map from $\{0,1\}^m$ to $\{0,1\}^{k_m}$.

Proposition 4.9 (Conditional converse)**.** *Let $L \in \mathsf{E}$ satisfy $\mathrm{C}(L_m) \geq 2^{\varepsilon m}$ for infinitely many m, and suppose that $\mathrm{C}(q_m) \leq 2^{\varepsilon m - 1}$ for all sufficiently large m. Then there are explicit exponents T_n such that $\mathrm{C}(P_n^{T_n}) \geq 2^{\varepsilon m - 1}$ for $n = 2k_m$ and infinitely many m. In particular, for every fixed $\varepsilon' < \varepsilon/2$, $\mathrm{C}(P_n^{T_n}) \geq 2^{\varepsilon' n}$ for infinitely many even n.*

Proof. The map $m \mapsto k_m$ is strictly increasing. For $n = 2k_m$, let T_n be the least nonnegative integer with $T_n \equiv L(z) \pmod{q_m(z)}$ for every $z \in \{0,1\}^m$, and put $T_n = 0$ for all other even n. The moduli are distinct primes, so the Chinese remainder theorem gives T_n. It is computable in time $2^{O(m)} = 2^{O(n)}$ from the truth table of L_m and the 2^m primes, so (T_n) is explicit.

Given a circuit for r_{T_n} on k_m-bit moduli, compose it with a circuit for q_m. On input z the result outputs $T_n \bmod q_m(z) = L(z) \in \{0,1\}$ in its low bit. Therefore

$$\mathrm{C}(P_n^{T_n}) \geq \mathrm{C}(r_{T_n}) \geq \mathrm{C}(L_m) - \mathrm{C}(q_m) \geq 2^{\varepsilon m - 1}$$

for the infinitely many large m in the hypothesis, by Theorem 4.1. Since $n = 2m + O(\log m)$, this is at least $2^{\varepsilon' n}$ for any fixed $\varepsilon' < \varepsilon/2$ and all large such m. □

Hardwiring the table of q_m costs about 2^m gates, which exceeds the complexity of every m-input function, so the hypothesis on q_m needs a genuine construction. Counting primes in blocks gives one at the square root of the table size. Write $\pi(x)$ for the number of primes at most x.

Lemma 4.10 (Index-to-prime circuits)**.** *For every $k \geq 2$ there is a circuit with k inputs and k outputs, of size $2^{k/2}\,\mathrm{poly}(k)$, that on every z with $0 \leq z < \pi(2^k - 1)$ returns the $(z+1)$st prime. Its outputs on larger z are unspecified.*

Proof. Primality of k-bit integers is decidable in deterministic time $\mathrm{poly}(k)$ [16], and a multitape machine running in time t has circuits of size $O(t \log t)$ [17]. Hence there is a circuit PRIME_k of size $\mathrm{poly}(k)$ deciding primality of k-bit integers. (Alternatively, Miller–Rabin testing with bases fixed by Adleman's argument [18] gives the same bound nonuniformly.)

Put $h = \lceil k/2 \rceil$, $\Lambda = 2^h$ and $J = 2^{k-h}$, and split $\{0, \ldots, 2^k - 1\}$ into the blocks $[j\Lambda, (j+1)\Lambda)$ for $0 \leq j < J$. Hardwire the counts $c_j = \pi(j\Lambda - 1)$ of primes below each block, so $c_0 = 0$. Fix $z < \pi(2^k - 1)$ and let q be the $(z+1)$st prime, the unique prime with exactly z primes below it. If $j\Lambda \leq q$, the primes below $j\Lambda$ are among the z primes below q, so $c_j \leq z$; if $j\Lambda > q$, they include q and the primes below it, so $c_j \geq z + 1$. Thus q lies in block j^*, the largest j with $c_j \leq z$.

The circuit computes the flags $f_j = [c_j \le z]$ with J comparisons. They are monotone, with $f_0 = 1$, so $g_j = f_j \wedge \neg f_{j+1}$ (with $f_J = 0$) is one exactly at j^*. An OR of masked words selects j^* and c_{j^*}, and a subtraction gives the rank $\rho = z - c_{j^*}$: the number of primes in block j^* below q. These steps use $O(Jk)$ gates.

For each offset $0 \le i < \Lambda$, a copy of PRIME_k evaluates $b_i = [j^*\Lambda + i \text{ is prime}]$; its input is j^* followed by the constant bits of i. A chain of $(h+1)$-bit incrementers computes $S_i = \sum_{i'<i} b_{i'}$. The selector $e_i = b_i \wedge [S_i = \rho]$ is one exactly at the offset of q. The output has high bits j^* and low bits the OR of the masked offsets $e_i \wedge i$. This part costs $\Lambda\,\mathrm{poly}(k) + O(\Lambda h)$ gates. Since J and Λ are at most $2 \cdot 2^{k/2}$, the total is $2^{k/2}\,\mathrm{poly}(k)$. □

Corollary 4.11 (Explicit hard powers above the square-root threshold)**.** *Let $L \in \mathsf{E}$ and $\varepsilon > 1/2$ satisfy $\mathrm{C}(L_m) \ge 2^{\varepsilon m}$ for infinitely many m. Then there are explicit exponents T_n such that, for infinitely many even n,*

$$\mathrm{C}(P_n^{T_n}) \ge 2^{\varepsilon n/2 - \varepsilon \log_2 n - O(1)}.$$

In particular $\mathrm{C}(P_n^{T_n}) \ge 2^{\varepsilon' n}$ infinitely often for every fixed $\varepsilon' < \varepsilon/2$, and ε' may be taken above $1/4$.

Proof. Pad the input of the circuit of Lemma 4.10, with $k = k_m$, by $k_m - m$ constant zero bits. Every $z \in \{0,1\}^m$ satisfies $z < 2^m \le \pi(2^{k_m} - 1)$ by the choice of k_m, so this computes q_m and

$$\mathrm{C}(q_m) \le 2^{k_m/2}\,\mathrm{poly}(k_m) = 2^{m/2}\,\mathrm{poly}(m),$$

using $k_m = m + \log_2 m + O(1)$. Since $\varepsilon > 1/2$, this is at most $2^{\varepsilon m - 1}$ for all large m. Proposition 4.9 then gives explicit T_n with $\mathrm{C}(P_n^{T_n}) \ge 2^{\varepsilon m - 1}$ for $n = 2k_m$ and infinitely many m. For such n, $m = n/2 - \log_2 m - O(1) \ge n/2 - \log_2 n - O(1)$, so $\varepsilon m - 1 \ge \varepsilon n/2 - \varepsilon \log_2 n - O(1)$. □

The full converse remains open only for $\varepsilon \le 1/2$. The Impagliazzo–Wigderson hypothesis asserts some fixed $\varepsilon > 0$, possibly small. Reaching that range by this route needs index-to-prime circuits of size $2^{\varepsilon m - 1}$ for every $\varepsilon > 0$, that is $2^{o(m)}$, which is far below the $2^{m/2}$ of Lemma 4.10. No such circuits are known: uniform deterministic generation of even one prime of a given length is open [15]. The choice of the $(z+1)$st prime is not the issue. By Theorem 4.1, the information in P_n^T is $T \bmod D_B$, which is determined by the residues of T modulo the prime powers up to B, each carrying at most k bits. An encoding that places 2^m independent bits in these residues must use about $2^m/k$ distinct prime powers. A reduction that queries r_T once per input must then produce suitable moduli from the index by a small circuit. Primes in arithmetic progressions, obtained from Linnik's theorem, lead to the same index-to-prime problem. Equivalently, the converse holds if some language in E remains exponentially hard when restricted to prime inputs, since then one may set $T \bmod p$ to that language at every prime $p \le B$; but this restates the problem rather than reducing it to the standard hypothesis. The Lean development does not formalize uniform complexity classes, so the results of this subsection are outside its checked boundary.

4.4 Short descriptions of hard exponents

Long binary expansions do not mean that hard exponents lack short descriptions. This subsection separates the length of an exponent from the complexity of describing it, and identifies the resource that matters. Its results are prose results; they are outside the Lean formalization of Section 8.

Fix a universal Turing machine U. For integers $T, n \ge 0$, the conditional Kolmogorov complexity $K(T \mid n)$ is the length of a shortest program p such that U, given p and the binary expansion of n, halts with the binary expansion of T on its output tape [19]. A different universal machine changes K by an additive constant only.

Proposition 4.12 (Hard exponents of constant description length)**.** *There is a constant c_U such that, for all sufficiently large even n, some exponent $T_n^* < D_B$ satisfies*

$$K(T_n^* \mid n) \le c_U \qquad \text{and} \qquad \mathrm{C}(P_n^{T_n^*}) \ge c\,\frac{2^{n/2}}{n},$$

where $c > 0$ is the constant of Theorem 4.5. Consequently no bound of the form $\mathrm{C}(F^T) \le \beta(n, \mathrm{C}(F), K(T \mid n))$ with β a polynomial holds, and any valid bound of this form satisfies $\beta(n, O(n), O(1)) = \Omega(2^{n/2}/n)$. The same holds for the unconditional complexity, since $K(T_n^*) \le 2\log_2 n + O(1)$.

Proof. Decreasing c if necessary, take c rational. Let $\theta_n = \left\lceil c2^{n/2}/n \right\rceil$, and let T_n^* be the least $T \ge 0$ with $\mathrm{C}(P_n^T) \ge \theta_n$. By Theorem 4.5 such a T exists below D_B for all sufficiently large even n. The following fixed program computes T_n^* from n. For $T = 0, 1, 2, \ldots$ in turn, compute the truth table of P_n^T by iterating Equation (4.1) on each of the 2^n states. Then enumerate every circuit with fewer than θ_n gates and evaluate it on all 2^n inputs. Output the first T for which no such circuit agrees with the truth table. Each stage is a finite computation, and the search stops at T_n^*. Let c_U be the length of this program. For the unconditional bound, prepend a self-delimiting encoding of n. Since $\mathrm{C}(P_n) = O(n)$, these exponents defeat every polynomial β. □

By Corollary 4.7, the exponent T_n^* still has $\Omega(2^{n/2}/n^2)$ binary digits. So hard exponents are long but can be simple. The upper bound of Theorem 1.1 does not depend on T at all, so in terms of description complexity the two results match up to $2^{O(\log n)}$ factors.

The program in the proof is expensive. It tries up to D_B exponents, each against $2^{O(\theta_n \log \theta_n)}$ circuits, so it runs in time $2^{2^{O(n)}}$. It uses only space $2^{O(n)}$: at any moment it stores one exponent below D_B, one candidate circuit, and one truth table. We do not claim that a substantially faster short program for a hard exponent cannot exist. The next statements show what a cheaper program would imply.

For a family of exponents T_n, a *description in time $t(n)$* (respectively *in space $S(n)$*) is a family of programs p_n of length $\mathrm{poly}(n)$ such that U, given p_n and n, outputs T_n within $t(n)$ steps (respectively, halts using at most $S(n)$ cells of work tape; the write-only output tape is not charged). For a family of maps, membership in $\mathsf{PSPACE}/\mathrm{poly}$ or $\mathsf{EXP}/\mathrm{poly}$ refers to the language of pairs (x, i) such that bit i of the image of x is one, with polynomial-length advice for each n.

Definition 4.13 (Residue profile)**.** For $T \ge 0$ and $n \ge 1$, the *residue profile* $\rho_{T,n} : \{0, 1, \ldots, 2^n\} \to \{0, 1, \ldots, 2^n\}$ is given by $\rho_{T,n}(p) = T \bmod p$ for $1 \le p \le 2^n$, and $\rho_{T,n}(0) = \min(T, 2^n)$.

Proposition 4.14 (Residue profiles control powers)**.** *Let $F_n : \{0,1\}^n \to \{0,1\}^n$ have $\mathrm{C}(F_n) \le \mathrm{poly}(n)$, and let $T_n \ge 0$.*

(i) If the family $\rho_{T_n,n}$ is in $\mathsf{PSPACE}/\mathrm{poly}$, then so is the family $F_n^{T_n}$. The same holds with $\mathsf{EXP}/\mathrm{poly}$ in place of $\mathsf{PSPACE}/\mathrm{poly}$.

(ii) For the permutations of Equation (4.1), the family $P_n^{T_n}$ is in $\mathsf{PSPACE}/\mathrm{poly}$ if and only if the family of remainder maps r_{T_n} of Theorem 4.1 is.

Proof. (i) Write $N = 2^n$, $T = T_n$, $F = F_n$, and take as advice the circuit of F together with the advice for $\rho_{T,n}$. Given x, compute $y = F^N(x)$ by N evaluations of F with an $(n+1)$-bit counter. The state y lies on the cycle reached from x. Its cycle length λ is the least $j \ge 1$ with $F^j(y) = y$. The tail length $\mu < N$ is the least i with $F^i(x) = F^{i+\lambda}(x)$. Each of these searches stores $O(n)$ bits

besides one evaluation of F. Read $m = \rho_{T,n}(0)$. If $m < \mu$, then $T = m$, and the output is $F^m(x)$. Otherwise $T \geq \mu$, because $\mu < N$, and

$$F^T(x) = F^{\mu+(((T \bmod \lambda)-\mu) \bmod \lambda)}(x),$$

which uses the value $T \bmod \lambda = \rho_{T,n}(\lambda)$ and at most $\mu + \lambda \leq 2N$ further evaluations. Each query to $\rho_{T,n}$ is answered by recomputing the requested bits in polynomial space, so the whole computation runs in polynomial space with polynomial advice. The computation also runs in time $2^{O(n)}\,\mathrm{poly}(n)$ plus the cost of the queries, which gives the $\mathsf{EXP}/\,\mathrm{poly}$ statement.

(ii) The reductions in the proof of Theorem 4.1 are explicit polynomial-time computations in both directions: $r_T(a)$ is the second coordinate of $P_n^T(a, 0)$, and $P_n^T(a, b)$ is obtained from a, b, and $r_T(a)$ by one addition, one comparison, and one selection. □

Corollary 4.15 (Cheap descriptions give membership in a uniform class)**.** *Let F_n and T_n be as in Proposition 4.14.*

(i) If T_n has a description in space $\mathrm{poly}(n)$, then $F_n^{T_n}$ is in $\mathsf{PSPACE}/\,\mathrm{poly}$. Hence, if $\mathsf{PSPACE} \subseteq \mathsf{P}/\,\mathrm{poly}$, then $\mathrm{C}(F_n^{T_n}) \leq \mathrm{poly}(n)$.

(ii) If T_n has a description in time $2^{\mathrm{poly}(n)}$, then $F_n^{T_n}$ is in $\mathsf{EXP}/\,\mathrm{poly}$. Hence, if $\mathsf{EXP} \subseteq \mathsf{P}/\,\mathrm{poly}$, then $\mathrm{C}(F_n^{T_n}) \leq \mathrm{poly}(n)$.

Proof. (i) A description in space $S = \mathrm{poly}(n)$ halts within $2^{O(S)}$ steps, so it emits at most $2^{O(S)}$ output bits. To compute $T \bmod p$ for $p \leq 2^n$, rerun the program and fold each output bit into a running remainder as it appears. For most-significant-first output, replace r by $2r + \text{bit} \bmod p$. For least-significant-first output, also maintain $2^j \bmod p$. A running value capped at 2^n gives $\min(T, 2^n)$ in the same way: for least-significant-first output, it suffices to record whether a one appears at position n or later. This uses $S + O(n)$ cells, so $\rho_{T_n,n}$ is in $\mathsf{PSPACE}/\,\mathrm{poly}$ with the program as advice, and Proposition 4.14 applies. (ii) The same streaming computation runs in time $2^{\mathrm{poly}(n)}$. In both parts, the final implication follows by folding the polynomial advice into the input: the resulting language lies in PSPACE (respectively EXP), so it has polynomial-size circuits under the stated inclusion, and fixing the advice gives circuits for $F_n^{T_n}$. □

So among short descriptions, space rather than length or time is the relevant resource. Constant-length descriptions in space $2^{O(n)}$ reach hard exponents by Proposition 4.12. Polynomial-space descriptions reach a hard power only if $\mathsf{PSPACE} \not\subseteq \mathsf{P}/\,\mathrm{poly}$. Time-$2^{\mathrm{poly}(n)}$ descriptions reach one only if $\mathsf{EXP} \not\subseteq \mathsf{P}/\,\mathrm{poly}$.

Long but easy exponents. Several exponents with exponentially long expansions have residue profiles in PSPACE, so by Proposition 4.14 their powers of any polynomial-size map are in $\mathsf{PSPACE}/\,\mathrm{poly}$:

- $T = 2^{2^n}$: compute $2^{2^n} \bmod p$ by 2^n modular squarings with an n-bit counter.
- Power towers $T = 2 \uparrow\uparrow h$ of any height $h < 2^{\mathrm{poly}(n)}$, with h given as advice: reduce exponents modulo iterated totients, using $a^b \equiv a^{(b \bmod \varphi(p))+\varphi(p)} \pmod p$ for $b \geq \log_2 p$. Whenever an intermediate exponent $2 \uparrow\uparrow j$ is below n, it is computed exactly and used directly, so the condition is needed only where it holds. The iterated totient of $p \leq 2^n$ reaches 1 after $O(n)$ steps, so only $O(n)$ levels need explicit work, and each totient is found by trial division in polynomial space.

- $T = D_B - 1$: $D_B \bmod p$ is a product of the largest prime powers not exceeding B, enumerated in polynomial space.

The last example is the inversion exponent of P_n, so $P_n^{D_B-1} = P_n^{-1}$. As noted in Section 3.6, this power even has a linear-size circuit. Membership in $\mathsf{PSPACE}/\operatorname{poly}$ gives only an upper bound. These examples show that neither the length of an exponent nor its numerical size forces hardness.

5 Input times and shared evaluation

5.1 An iteration count supplied as input

Definition 5.1. For $m \geq 1$, define

$$\mathrm{Iter}_{F,m} : \{0,1\}^n \times \{0,1\}^m \longrightarrow \{0,1\}^n, \qquad \mathrm{Iter}_{F,m}(x,t) = F^{\mathrm{val}(t)}(x),$$

where $\mathrm{val}(t)$ is the nonnegative integer encoded by t.

Theorem 5.2 (Variable-time iteration)**.** *Under the assumptions of Theorem 1.1, for every $m \geq 1$,*

$$\mathrm{C}(\mathrm{Iter}_{F,m}) = O\Big(m\left(A + \sqrt{n^3 NA}\right)\Big). \tag{5.1}$$

In particular,

$$\mathrm{C}(\mathrm{Iter}_{F,n}) = O\Big(n\left(A + \sqrt{n^3 NA}\right)\Big).$$

If s and m are polynomial in n, then $\mathrm{C}(\mathrm{Iter}_{F,m}) \leq 2^{n/2+O(\log n)}$.

Proof. Write $t = \sum_{j=0}^{m-1} t_j 2^j$. For each j, apply Theorem 1.1 directly to the original map F and the fixed exponent 2^j, obtaining a circuit for $G_j = F^{2^j}$. Starting with $v_0 = x$, use

$$v_{j+1} = \begin{cases} G_j(v_j), & t_j = 1, \\ v_j, & t_j = 0. \end{cases}$$

Inductively, $v_j = F^{\sum_{a<j} t_a 2^a}(x)$, so $v_m = F^t(x)$. There are m circuits of the size in Equation (1.2) and m n-bit multiplexers. The extra $O(mn)$ gates are absorbed by the claimed bound. Each G_j is constructed from F using the theorem; it is not obtained by duplicating the circuit for G_{j-1}. □

Theorem 5.3 (Long input times)**.** *Write $B = A + \sqrt{n^3 NA}$. For every $m \geq 1$,*

$$\mathrm{C}(\mathrm{Iter}_{F,m}) = O(\min\{mB,\ nB + nm\}).$$

Thus the cost of processing additional time bits is additive once the state width is fixed.

Proof. The first bound is Theorem 5.2. For the second, compute the exact eventual cycle length $\ell = \ell(x)$ in $O(B)$ gates using Theorem 6.1. Replace the input time by

$$u = \begin{cases} t, & t < N, \\ N + (t - N) \bmod \ell, & t \geq N. \end{cases}$$

Then $u < 2N$ and $F^u(x) = F^t(x)$: in the second case both times are past cycle entry and have the same residue modulo ℓ.

Compute $t \bmod \ell$ by streaming its bits from most significant to least significant, retaining an $O(n)$-bit remainder. Each step doubles the remainder, adds one bit, and conditionally subtracts ℓ; a spare bit prevents overflow. Shared-carry addition and subtraction give $O(n)$ gates per input bit, including comparison and selection. The bounded modular correction by N and the final selection cost $O(n^3 + m)$ with the arithmetic circuits already used in the proof. Hence forming u costs $O(nm + n^3)$, and its $n + 1$ bits feed Theorem 5.2 at cost $O(nB)$. Since $n^3 \le B$, the second bound follows. If $m \le n$, one may simply use the original m-stage construction. □

5.2 Sharing a dictionary among independent trajectories

Let

$$B_{F,T,r}(x_1, \dots, x_r) = (F^T(x_1), \dots, F^T(x_r)).$$

The inputs $x_1, \dots, x_r$ are independent n-bit words. The following construction uses one checkpoint dictionary for all of them.

Theorem 5.4 (Batched fixed powers)**.** *Under the assumptions of Theorem 1.1, for every* $1 \le r \le N$, $T \ge 0$, *and* $1 \le L \le N$,

$$\mathrm{C}(B_{F,T,r}) = O\Big(rLs + n^3\left(rL + \frac{N}{L}\right)\Big). \tag{5.2}$$

Consequently,

$$\mathrm{C}(B_{F,T,r}) = O\Big(rA + \sqrt{rn^3NA}\Big). \tag{5.3}$$

Proof. Set $K = 2L$. If $T < K$, unroll T copies of F independently for each input. Otherwise use the checkpoint set and dictionary from Section 3. Generate the first $K + 1$ states of all r trajectories. Together with one copy of the dictionary, these give

$$R = |H| + r(K + 1) = O(N/L + rL)$$

records. Attach the trajectory number and within-trajectory position to each query. Because $r \le N$ and $L \le N$, the record width and $\log(R + 2)$ are $O(n)$. One application of Lemma 2.1 therefore resolves all queries in $O(n^3(N/L + rL))$ gates and returns them in their original order.

Select a matching checkpoint separately within each trajectory. Run repeated-state detection separately on each trajectory as well; this costs $O(rn^3L)$ gates and does not confuse a collision between two different trajectories with a repetition along one trajectory. The correction trajectory and output selection are also performed separately for each input. The fixed residue table may be implemented with a separate $O(nL)$-gate lookup for each trajectory, whose total cost is absorbed by $O(rn^3L)$. The two simulation phases together cost $O(rLs)$ gates. This proves Equation (5.2).

Choose

$$L = \max\left\{1, \left\lceil \sqrt{n^3N/(rA)} \right\rceil \right\}.$$

As before $1 \le L \le N$. If $rA \le n^3N$, then $rAL = O(\sqrt{rn^3NA})$ and $n^3N/L = O(\sqrt{rn^3NA})$. If $rA > n^3N$, the choice is $L = 1$ and the cost is $O(rA)$. This gives Equation (5.3). □

Corollary 5.5 (Independent input times)**.** *For* $1 \le r \le N$ *and* $m \ge 1$, *the map*

$$((x_1, t_1), \dots, (x_r, t_r)) \longmapsto (F^{t_1}(x_1), \dots, F^{t_r}(x_r)), \qquad t_i \in \{0,1\}^m,$$

has a circuit of size

$$O\Big(m\left(rA + \sqrt{rn^3NA}\right)\Big). \tag{5.4}$$

Proof. Use the binary-power construction of Theorem 5.2. At stage j, apply Theorem 5.4 with update map F and exponent 2^j on the r current states, then use the jth bit of each t_i to select whether to retain or update that state. The $O(mrn)$ selection gates are absorbed by Equation (5.4). □

For polynomial s and m, and $r = \lfloor 2^{\alpha n} \rfloor$ with fixed $0 \le \alpha \le 1$, Equation (5.4) is at most

$$2^{(1+\alpha)n/2+O(\log n)}.$$

This is an upper-bound sharing statement. It is not a direct-sum separation against optimal circuits for a single trajectory.

5.3 A polynomial improvement with AKS sorting networks

The factor n^3 in Theorem 1.1 has two sources: $O(n)$ gates per comparator on $O(n)$-bit records, and the $O(\log^2 R) = O(n^2)$ comparators per record of Batcher's networks. Ajtai, Komlós, and Szemerédi constructed comparator networks sorting R keys with $O(R \log R)$ comparators [2]. Replacing Batcher's networks by theirs saves one factor of n. Their constants are very large, so this is an asymptotic statement only. The Lean development mechanizes the Batcher version; nothing in this subsection is mechanized.

Lemma 5.6 (Lookup and repetition with AKS networks)**.** *The dictionary lookup of Lemma 2.1 and the repeated-state detection of Lemma 2.2 have circuits of sizes*

$$O\big((M+Q)(w+\log(M+Q+2))\log(M+Q+2)\big) \quad \textit{and} \quad O\big(Q(w+\log(Q+2))\log(Q+2)\big)\,,$$

respectively. In particular, for $w = O(n)$ and $\log(M+Q+2) = O(n)$ both costs are $O(n^2(M+Q))$.

Proof. Repeat the proofs of Lemmas 2.1 and 2.2 with every sorting network replaced by an AKS network on the same records, padded with dummy records as before. The networks use $O(R\log(R+2))$ comparators, each of cost $O(w+\log(R+2))$. The scans and the selection of a flagged record cost $O(R(w+\log(R+2)))$ gates and are unchanged. □

Theorem 5.7 (AKS-strengthened tradeoff)**.** *Under the assumptions of Theorem 1.1, put $A_2 = s+n^2$. For every integer $T \ge 0$ and every integer $1 \le L \le N$,*

$$\mathrm{C}(F^T) = O\Big(Ls + n^2\left(L + \frac{N}{L}\right)\Big), \qquad \textit{hence} \qquad \mathrm{C}(F^T) = O\Big(A_2 + n\sqrt{NA_2}\Big)\,.$$

Proof. Use the construction of Section 3 unchanged, and implement its lookup and repeated-state detection by Lemma 5.6. These now cost $O(n^2(N/L+L))$ and $O(n^2L)$ gates. Every other component already fits this bound: the two simulations cost $O(Ls)$; the residue-table multiplexer, the trajectory selections, and the final output selection cost $O(nL)$; and the long division and modular correction of Equation (3.4) cost $O(n^2)$. For the optimized bound, choose $L = \max\{1, \lceil\sqrt{n^2N/A_2}\rceil\}$ and argue as for Equation (3.6), with n^3 replaced by n^2 and A by A_2. Then $L \le \lceil\sqrt{N}\rceil \le N$ since $A_2 \ge n^2$, and $\sqrt{n^2NA_2} = n\sqrt{NA_2}$. □

The proofs of the following results use the lookup and repetition circuits only through the per-record cost of Lemmas 2.1 and 2.2, together with $O(Ls)$ simulation, $O(nL)$ selection, and $O(n^2)$ arithmetic. Each therefore holds with n^3 replaced by n^2 in its parameterized bound and with A replaced by A_2 in its optimized bound: the long-trajectory, invariant-set and image-sensitive bounds (Theorems 3.2 and 3.3 and Corollary 3.4), exact permutation inversion (Corollary 3.5),

variable-time iteration (Theorem 5.2, by m applications of Theorem 5.7), and batched fixed powers and independent input times (Theorem 5.4 and Corollary 5.5). For the batched bound, the shared lookup costs $O(n^2(N/L + rL))$ and the r separate repetition tests cost $O(rn^2L)$, giving

$$\mathrm{C}(B_{F,T,r}) = O\Big(rA_2 + n\sqrt{rNA_2}\Big).$$

For the permutation of Section 4, $\mathrm{C}(P_n) = O(n)$ gives $A_2 = O(n^2)$ and $\max_T \mathrm{C}(P_n^T) = O(n^2 2^{n/2})$, within a factor $O(n^3)$ of the lower bound in Theorem 4.4. We do not claim the improvement for the navigation and target results of Section 6 or for Theorem 5.3, whose proofs contain further n^3 terms that we have not re-examined.

6 Functional-graph navigation

Write $B = A + \sqrt{n^3NA}$. For a state x, let $\mu(x)$ be its distance to its eventual cycle, let $\ell(x)$ be that cycle's length, and let $c(x)$ be the lexicographically least state on the cycle. Define $\phi(x) \in \{0, \ldots, \ell(x) - 1\}$ by

$$F^t(x) = F^{(\phi(x)+t) \bmod \ell(x)}(c(x)) \qquad (t \geq \mu(x)). \tag{6.1}$$

The phase is an eventual-orbit label; it does not assert that a transient state itself lies on the cycle. These labels are unique. Cycle lengths are encoded with $n+1$ bits, since a cycle may have length N.

Theorem 6.1 (Cycle labels)**.** *A circuit of size $O(B)$ computes $(\ell(x), c(x), \phi(x))$ for every x.*

Proof. Fix L and use the checkpoint construction with $K = 2L$. Hardwire the three labels for each checkpoint. If $F^d(x) = h$ for $d \leq K$, output

$$\ell(x) = \ell(h), \qquad c(x) = c(h), \qquad \phi(x) = (\phi(h) - d) \bmod \ell(h).$$

This follows by applying Equation (6.1) after both transients have ended. All records still have $O(n)$ bits.

If there is no checkpoint hit, the prefix repeats. Sort prefix records by state and then by original index. Among adjacent equal-state pairs, choose the pair with smallest second index j, retaining its first index i. This is the first repetition in the original trajectory: any earliest second occurrence is adjacent to its first occurrence in its equal-state run. The states before j are distinct, so $i = \mu(x)$ and $[i, j)$ consists of exactly one cycle traversal. Set $\ell = j - i$, and select the minimum state $c = x_p$ over $i \leq p < j$. Then $\phi(x) = (-p) \bmod \ell$. Minimum selection and the interval mask cost $O(nK)$ gates, and word arithmetic costs $O(n^2)$ gates. Sorting and lookup cost $O(n^3(L + N/L))$, while the prefix costs $O(Ls)$. Optimizing L proves the claim. Discarded branches use safe positive dummy moduli, as in the fixed-power construction. □

Theorem 6.2 (Transient depth)**.** *The exact transient depth $\mu(x)$ is computable by a circuit of size $O(B)$.*

Proof. Put $D = N!$ and $R = F^D$. Every trajectory is cyclic after N steps, every cycle length divides D, and $D \geq N$. Thus R maps into the cyclic states, fixes each such state, and satisfies $R^2 = R$. It commutes with all iterates of F, so

$$F^i(x) = F^i(R(x)) \quad \Longleftrightarrow \quad i \geq \mu(x). \tag{6.2}$$

Indeed, equality is equivalent to $F^i(x) = R(F^i(x))$, which is exactly cyclicity. The projection costs $O(B)$ gates by Theorem 1.1.

Fix a checkpoint parameter L and set $K = 2L$. Retain the first $K+1$ states of both trajectories and compare corresponding states. If the states at position K agree, select the first equality; its index is $\mu(x)$. Otherwise $\mu(x) > K$, so the original prefix has no repetition and must meet a checkpoint $h = F^d(x)$ with $d < K$. Every such hit is transient and satisfies $\mu(x) = d + \mu(h)$.

Hardwire $\mu(h)$ with each checkpoint and resolve all prefix queries using Lemma 2.1. Either branch uses $O(n)$-bit records; comparing, selecting, and adding indices cost $O(nL)$ gates. The two prefixes and shared lookup cost $O(Ls + n^3(L + N/L))$. Use the same L for the projection and these operations, then optimize as in Equation (3.6). □

Lemma 6.3 (Compressed checkpoint forest)**.** *Let H be the checkpoint set from Lemma 2.3. Cut the outgoing edge of every cyclic vertex, and let S be the union of the paths from H to their first cyclic vertices, including both endpoints. There is a set J with $H \subseteq J \subseteq S$ and $|J| \le 2|H|$ such that:*

1. *Every root and every branching vertex of the forest on S lies in J.*
2. *From each nonroot $a \in J$, the next vertex of J is reached after some $1 \le d_a \le 2L$ steps. The segments $a, F(a), \ldots, F^{d_a-1}(a)$ are pairwise disjoint.*
3. *Each $a \in J$ has labels $p(a), q(a) < N$ such that, for $a, h \in J$,*

$$p(a) \le p(h) \le q(a) \quad \iff \quad F^t(h) = a \text{ for some } 0 \le t \le \mu(h).$$

Proof. Take J to be H together with the roots and branching vertices of S. Every leaf belongs to H: any other vertex on a checkpoint path has a predecessor on that path. In a rooted forest, the number of roots plus the number of vertices with at least two children is at most the number of leaves. Indeed, summing the numbers of children gives the number of nonroot vertices; a branching vertex contributes at least two. Therefore $|J| \le |H| + |\text{leaves}| \le 2|H|$.

Apply checkpoint coverage to $F(a)$. A checkpoint hit gives a positive hit from a within $2L$ steps; a repetition gives cycle entry within L steps, where a root is marked. Hence the next marked vertex is at distance at most $2L$. Between marks, each vertex of S has a unique predecessor in S, so two such segments cannot merge. Finally order vertices by a preorder traversal of the cut forest. Descendants of a vertex occupy a contiguous interval; its first and last ranks give p, q. Only the labels of J need to be stored. The Lean proof obtains the same interval property by lexicographically ordering root-to-vertex paths, without formalizing a traversal algorithm. □

Theorem 6.4 (Reachability and first hitting times)**.** *A circuit of size $O(B)$, on input (x, y), decides whether $F^t(x) = y$ for some $t \ge 0$ and returns the least such t when it exists. The output may use a separate existence flag and n time bits.*

Proof. Use a common parameter L in the depth and cycle-label constructions. Write $K = 2L$ and construct the forest of Lemma 6.3. Store a dictionary of the marked vertices $a \in J$, with their states, depths, and interval labels. A second dictionary has one key $F^{d_a-1}(a)$ for each nonroot mark a, with the same payload for a. These last-predecessor keys are distinct by disjointness of the segments. Both dictionaries have $O(N/L)$ records of $O(n)$ bits.

Suppose first that y is transient. Generate the prefixes of x and y through time K. If y occurs in the prefix of x, accept. Otherwise select any marked vertex h in that prefix; if none exists, reject. To select a candidate ancestor a, use $a = y$ if $y \in J$. Otherwise query the second dictionary on the

prefix of y, retain only matches with $\mu(a) \geq \mu(y)$, and select any retained record. Reject if there is no such record. Finally accept exactly when

$$y \in \{a, F(a), \ldots, F^K(a)\} \quad \text{and} \quad p(a) \leq p(h) \leq q(a).$$

The first test uses a third short prefix. It is essential for rejecting side branches outside S.

Every acceptance is sound: x reaches h, the interval test says that h reaches a, and the verification prefix reaches y. Conversely, suppose x reaches a transient y beyond its inspected prefix. That prefix cannot repeat and must contain a checkpoint. Every selected mark h precedes y, so $y \in S$. If y is unmarked, it lies in a unique compressed segment, whose last-predecessor key appears within K steps of y. Its lower endpoint passes the depth filter. Moreover, every retained key has the same property: if $F^j(y) = F^{d_a-1}(a)$ and $\mu(a) \geq \mu(y)$, decreasing depth gives $j \leq d_a - 1$. Unique predecessors inside the segment then give $y = F^{d_a-1-j}(a)$. A marked h can reach this segment only through a. Thus the selected record passes both final tests. The marked-target case takes $a = y$ directly.

Three prefixes cost $O(Ls)$ gates. Shared lookup costs $O(n^3(L + N/L))$, and depth filtering, record selection, prefix equality, and interval comparisons add $O(nL + n^3)$ gates. Depths and cycle labels fit the same tradeoff. Optimization therefore gives $O(B)$ gates. For a positive transient query, the first hitting time is simply $\mu(x) - \mu(y)$, since depth decreases by one on each transient step.

If y is cyclic, it is reachable exactly when $c(x) = c(y)$. On a positive instance its first hitting time is

$$\mu(x) + \big((\phi(y) - \phi(x) - \mu(x)) \bmod \ell(x)\big),$$

where the parenthesized residue lies in $[0, \ell(x) - 1]$. This is the first time on or after entry into the cycle with the required phase. Finally $\mu(x) + \ell(x) \leq N$, so every first hitting time is at most $N - 1$. Arithmetic may use $n + 1$ intermediate bits. The time field is unspecified on negative instances; an additional $O(n)$-gate mask can set it to zero if desired. □

Corollary 6.5 (Complete visit schedules)**.** *A circuit of size $O(B)$, on input (x, y), returns a success flag and two integers d, p describing every visit time. A zero flag denotes no visits. For a positive flag, $d < N$ is the first visit, $0 \leq p \leq N$, and*

$$\{t \geq 0 : F^t(x) = y\} = \{d + kp : k \geq 0\}.$$

Here $p = 0$ for transient y, giving the singleton $\{d\}$, and $p = \ell(y)$ for cyclic y. The period field uses $n + 1$ bits.

Proof. Use Theorem 6.4 for the flag and d, and Theorem 6.1 and an $O(B)$ cyclicity test for p. After the first visit, a later time $t \geq d$ is a visit exactly when $F^{t-d}(y) = y$. Return times of a cyclic point are precisely the multiples of its least positive period. A transient point has no positive return. This proves the descriptor. □

For example, a visit occurs by deadline T exactly when the flag is positive and $d \leq T$. The number of visits at times $0, \ldots, T$ is

$$\begin{cases} 0, & \text{flag zero or } T < d, \\ 1, & \text{flag positive, } T \geq d,\ p = 0, \\ 1 + \lfloor (T - d)/p \rfloor, & \text{flag positive, } T \geq d,\ p > 0. \end{cases}$$

These are arithmetic consequences of the returned descriptor; the descriptor bound does not include a circuit for processing an additional deadline input of unspecified length.

Relation to navigation data structures. Function powers and path queries have established data-structure formulations [4, 8]. In particular, nearest labeled successors include reachability of a fixed singleton target. Those results charge representation space and query time in a RAM model. Here the given object is a succinct transition circuit, and every gate accessing a nonuniform table is charged. These differences specify the claim being compared; they do not establish historical novelty of the present bounds.

6.1 Recognizable targets and terminating computations

Corollary 6.6 (First recognizable target)**.** *Let $P : \{0,1\}^n \to \{0,1\}$ have circuit size a, and put $A_P = s + a + n^3$. Eventual target membership,*

$$H(x) = 1 \quad \Longleftrightarrow \quad \exists t \geq 0 : \ P(F^t(x)) = 1,$$

has a circuit of size $O(A_P + \sqrt{n^3 N A_P})$. Within the same bound, a circuit returns the first visited target state and its exact first visit time on positive instances.

Proof. Set $G(x) = x$ when $P(x) = 1$ and $G(x) = F(x)$ otherwise. This costs $O(s + a + n)$ gates. A first target, if one exists, is visited before step N, since an earlier repeated state would preclude a later first visit. Hence $G^N(x)$ is the first target state on positive instances, and $H(x) = P(G^N(x))$. Apply Theorem 1.1 to G. For the time, apply Theorem 6.2 to G. On a positive instance, the first target is an absorbing fixed point. A cyclic state cannot precede it on the trajectory: iteration is injective on cyclic states, so any cyclic predecessor of that fixed point is the same point. Consequently $\mu_G(x)$ is exactly the first target time. □

Corollary 6.7 (Complete-configuration simulation)**.** *Suppose a deterministic computation has its entire configuration encoded in b bits and a next-configuration circuit of size s. If halting configurations are absorbing and every valid initial configuration halts, its final-configuration map on valid initial states has circuit size*

$$O\left(s + b^3 + \sqrt{b^3 2^b (s + b^3)}\right).$$

The exact first halting time can also be returned within this bound.

Proof. Before halting, no configuration can repeat. The iterate at time 2^b is therefore the final configuration. Apply Theorem 1.1 with state width b. The first halt is the transient depth, by the same absorbing-state argument as in Corollary 6.6; use Theorem 6.2. Initialization and output decoding circuits are charged separately. □

In particular, when $s = \mathrm{poly}(b)$, both the final configuration and the first halting time have circuit size

$$2^{b/2}\,\mathrm{poly}(b).$$

For a nearly in-place computation on an n-bit input, with complete state width $b = n + O(\log n)$ and a polynomial-size transition, this specializes to $2^{n/2}\,\mathrm{poly}(n)$. Polynomial-size initialization and output decoding preserve that bound. The parameter b includes all persistent input information, control, and workspace; these bits are counted even when an abstract machine model calls part of its storage read-only. A general polynomial-space machine can have far more than $n + O(\log n)$ configuration bits, so the corollary gives no such input-length bound for all of PSPACE.

6.2 Synchronous and recurring queries

The projection $R = F^{N!}$ from the proof of Theorem 6.2 also supplies queries about the eventual dynamics.

Corollary 6.8 (Eventual synchronous coalescence)**.** *A circuit of size $O(B)$ decides, on input (x, y), whether there exists $t \geq 0$ such that $F^t(x) = F^t(y)$.*

Proof. The required equivalence is

$$\exists t \geq 0 : \ F^t(x) = F^t(y) \quad \Longleftrightarrow \quad F^N(x) = F^N(y).$$

Equality before time N persists to time N. Both states are cyclic by that time, and F restricts to a bijection on the cyclic states. Therefore later equality can be canceled back to time N. Compute two copies of F^N and compare their outputs. □

This query requires the same elapsed time on both trajectories. Orbit intersection permits different elapsed times and is a different predicate; for example, distinct points of one cycle never coalesce synchronously.

Corollary 6.9 (Recurring and permanent targets)**.** *For a fixed predicate $P : \{0,1\}^n \to \{0,1\}$ of circuit size a, each of*

$$\forall u \ \exists t \geq u : \ P(F^t(x)) = 1, \qquad \exists u \ \forall t \geq u : \ P(F^t(x)) = 1$$

is decidable by a circuit of size $O(A_P + \sqrt{n^3 N A_P})$, where $A_P = s + a + n^3$.

Proof. A target recurs arbitrarily late exactly when the eventual cycle contains an accepting state. Start at $R(x)$ and use the absorbing-target construction in Corollary 6.6. If G makes accepting states absorbing, the circuit outputs $P(G^N(R(x)))$. A successful visit is periodic and hence recurs arbitrarily late; every sufficiently late visit from x is a future visit from $R(x)$. The projection and the target circuit each obey the stated bound. Permanent eventual acceptance is the complement of recurrence of the predicate $1 - P$, which changes the bound only by a constant number of gates. □

7 Structured transitions

The universal bound uses only the state width and the size of one transition. Additional structure can give compact circuits for whole iterates even when an orbit visits exponentially many states. Affine payload updates stay affine along each autonomous controller trajectory; sparse counter edits give orbit coordinates described by a few intervals. In both cases, selecting and evaluating the resulting description has a charged circuit cost that can improve on the universal bound.

7.1 Small controllers with affine payloads

The next result isolates all exponential cost in the autonomous controller, allowing a much larger affine payload.

Theorem 7.1 (Affine-controller powering)**.** *Let $q + p = n \geq 1$, and write a state as $(u, v) \in \{0,1\}^q \times \mathbb{F}_2^p$. Suppose*

$$F(u, v) = \big(g(u), A(u)v + b(u)\big),$$

where $g : \{0,1\}^q \to \{0,1\}^q$, $A(u)$ is a $p \times p$ binary matrix, and $b(u) \in \mathbb{F}_2^p$. For every fixed integer $T \geq 0$,

$$\mathrm{C}(F^T) = O\Big(2^q(q + p^2)\Big).$$

For an m-bit input exponent, the size bound is $O(m2^q(q + p^2))$. In particular, $q = O(\log n)$ gives polynomial-size circuits for every fixed power, uniformly in the magnitude of T. The matrices may be singular and need not commute.

Proof. Once the initial controller u is fixed, the controller trajectory and all affine maps applied to v are fixed. Composition of affine maps is affine, so for some hardwired matrix $M_T(u)$ and vector $d_T(u)$,

$$F^T(u, v) = \big(g^T(u), M_T(u)v + d_T(u)\big).$$

For each of the 2^q controller values, evaluate its fixed affine map using $O(p^2)$ gates and provide its fixed controller output. Compute each controller-equality flag once, and share it across the $q + p$ output bits. A scan selects the matching record using $O(2^q(q + p))$ additional gates. This proves the fixed-power bound, including a charge for every table-selection gate.

For an input exponent, conditionally apply the circuits for F^{2^j} to the current full state, as in Section 5.1. Each stage updates the controller as well as the payload and adds $O(q + p)$ selection gates. Summing over m stages gives the second bound. □

Autonomy is essential to this argument: if the controller update depends on v, its future choices need not be determined by the initial u. No efficient algorithm for obtaining $M_T(u)$ from a succinct transition is asserted here. The bound uses ordinary affine composition and finite selection; it does not claim a new algebraic closure principle.

The same reasoning applies beyond affine maps. Let $\mathcal{M}$ be a family of maps on p-bit payloads containing the identity and closed under composition, and suppose *every* member has a circuit of size at most κ. If the autonomous controller selects maps from $\mathcal{M}$, every fixed power costs

$$O(2^q(\kappa + q + p + 1)),$$

and an m-bit input time multiplies this by m. Closure alone is insufficient: the uniform evaluation bound on composite maps is a separate hypothesis. For nonuniform circuit existence, no efficient procedure for computing those composites is needed.

7.2 Sparse modifications of a counter

The state-space parameter is not the only source of compression. Identify $\{0,1\}^n$ with $\mathbb{Z}/N\mathbb{Z}$, where $N = 2^n$, and write $R(x) = x + 1 \pmod N$. A permutation obtained by changing a small number of counter successors admits a bound linear in its sparse description, even when it is a single cycle through all N states.

Theorem 7.2 (Sparse-counter powers)**.** *Let $n \geq 1$, let F be a permutation of $\mathbb{Z}/N\mathbb{Z}$, and let $r = |\{x : F(x) \neq R(x)\}|$. For every fixed $T \in \mathbb{Z}$,*

$$\mathrm{C}(F^T) = O\big(n(r+1)\big).$$

For an unsigned m-bit time supplied as input,

$$\mathrm{C}\big((x,t) \mapsto F^t(x)\big) = O\big(n(r+m+1)\big).$$

Both constants are absolute. The dependence on time-input length is additive.

7.2.1 Commutation defects and displacement

The fixed-power assertion follows from a more general statement.

Proposition 7.3 (Sparse-counter centralizers)**.** *Let F be any map on $\mathbb{Z}/N\mathbb{Z}$ differing from R at r inputs. Every permutation H commuting with F has a displacement function $\delta_H(x) = H(x) - x \pmod N$ that changes on at most $2r$ counter edges. Consequently, $\mathrm{C}(H) = O(n(r+1))$.*

Proof. Put $E = \{x : F(x) \neq R(x)\}$. If $x \notin E$ and $H(x) \notin E$, then

$$H(R(x)) = H(F(x)) = F(H(x)) = R(H(x)).$$

Thus the set of commutation failures for H and R lies in $E \cup H^{-1}(E)$ and has size at most $2r$. Commutation at x is exactly $\delta_H(x+1) = \delta_H(x)$. Cutting the ordinary linear order $0, \ldots, N-1$ at these changes gives at most $2r+1$ constant intervals. Compare the input with their endpoints, select a hardwired displacement, and add it modulo N, at cost $O(n(r+1))$.

An alternative implementation, used in Lean, avoids ordering the changes. Let $D = \{i : \delta_H(i+1) \neq \delta_H(i)\}$, with cyclic addition in the arguments and ordinary integer comparisons below. Telescoping gives

$$H(x) = x + \delta_H(0) + \sum_{\substack{i \in D \\ i < x}} \big(\delta_H(i+1) - \delta_H(i)\big) \pmod N.$$

Each summand uses one comparison, a conditional constant word, and a linear-size adder. The wraparound edge is never selected. All gates, including the shared carry chains, are charged. □

Every signed power of a permutation commutes with it, proving the first part of Theorem 7.2. The proposition only requires injectivity of H in its cardinality argument; it does not require F to be a permutation. Its set-containment argument works on arbitrary universes, before finiteness or coordinates are imposed.

The geometry relating commutation failures to blocks in cycles is established permutation theory. Moreno and Rivera characterize these blocks in their Theorem 3.8 and Corollary 3.10 [9]. Here we use the elementary defect containment and account for the Boolean cost of the resulting representation. Historical priority of these precise circuit bounds has not been determined.

7.2.2 A chart constructed from the edited successors

For the live-time bound, assume first $r > 0$ and cut the counter cycle immediately after each point of E. There are r nonempty circular arcs. Inside an arc F increments; its only possible change is at the last state. Since F and R are permutations agreeing off E,

$$F(E) = R(E).$$

Indeed, the images of the unedited inputs coincide, so their complements coincide. Therefore each edited edge leads to an arc start, and these edges induce a permutation of the r arcs.

Decompose that permutation into cycles and concatenate the arcs in each cycle, retaining their individual lengths. Assign to a state the triple (c, p, ℓ) consisting of its cycle identifier, its position within that cycle, and its total cycle length. The coordinate map and its inverse are piecewise translations on the original arcs and their concatenated images. At most one original arc crosses zero; splitting it there adds one linear interval. Comparisons, shared record selection, and addition therefore implement each direction in $O(n(r+1))$ gates. Identifiers and positions use n bits, while lengths use $n+1$ bits to allow $\ell = N$. All intermediate arithmetic uses $O(n)$ bits.

In these coordinates one step preserves c, ℓ and increments p modulo ℓ. For an m-bit input t, compute its remainder by scanning bits from most significant to least significant. With current remainder $u < \ell$, the next remainder is $2u + b$ or $2u + b - \ell$, according as $2u + b < \ell$ or not. A comparison and subtraction cost $O(n)$ per bit; one extra bit prevents overflow. Add p, reduce once more, and decode. The cost is $O(nr + nm + n)$. If $r = 0$, ordinary addition modulo N gives the same bound. This proves the second part of Theorem 7.2. For fixed signed T, hardwire T mod ℓ for each cycle instead.

Corollary 7.4 (Labels and navigation for sparse counters)**.** *For a permutation differing from the counter at r states, the cycle length, numerically least cycle vertex, and phase of an input state can be computed with $O(n(r+1))$ gates. The same bound suffices, on input (x, y), for reachability, the exact first hitting time, and a descriptor of all visits.*

Proof. Cut the ordinary linear order after each exception, obtaining at most $r+1$ intervals. Within each interval $[a, b)$, the trajectory follows ordinary addition. Hardwire its cycle representative c, cycle length ℓ, and phase p_a at its first state. An input x in that interval has label $(c, \ell, (p_a + x - a) \bmod \ell)$. The interval length is at most ℓ, so one addition and conditional subtraction suffice for the phase. Comparisons and record selection cost $O(n)$ per interval.

Compute the two labels. Since every state is cyclic, y is reachable from x exactly when $c(x) = c(y)$. On a positive instance, the first hit is

$$d = (\phi(y) - \phi(x)) \bmod \ell(x),$$

and all visits occur at times $d + k\ell(x)$, $k \geq 0$. Both phases are already below the common cycle length, so the difference needs only comparison and subtraction: use $\phi(y) - \phi(x)$ when nonnegative, and $\ell(x) - (\phi(x) - \phi(y))$ otherwise. This costs $O(n)$ additional gates. The descriptor also determines the exact bounded visit count; this is an arithmetic interpretation of its output, with no deadline input charged. □

Proposition 7.5 (Polynomial-time construction from an edit list)**.** *Given n and an explicit list of the r edited successors of the n-bit counter, one can check permutation preservation and construct an orbit chart of $O(r+1)$ records, each of $O(n)$ bits, in time polynomial in n and $r+1$. For a valid list, the chart supplies the encoding and decoding used above. Given also a signed binary integer T, one can construct a piecewise-translation table for F^T with at most $2r+1$ linear intervals in time polynomial in n, $r+1$, and $\log(|T|+2)$.*

Proof. Compare the edited outputs with the freed counter outputs to check $F(E) = R(E)$, after checking distinct input keys. Sort the cuts, compute arc lengths, follow the induced permutation of the arcs, and form prefix sums on each of its cycles. These operations construct the chart using arithmetic on $O(n)$-bit integers. The case $r = 0$ is the ordinary counter chart. For a signed power, reduce T modulo each chart cycle length. Use the chart to evaluate F^{-T} on E, and test the $O(r)$

candidate change edges $E \cup F^{-T}(E)$ from Proposition 7.3. Sorting their successors and evaluating the displacement at the interval starts gives the table; discard boundaries without a displacement change. □

The input here is the explicit edit list, whose entries specify where counter edges have changed and their new successors. Given only a circuit for F, identifying that list is a separate task: the proposition asserts no polynomial-time extraction of exceptions from a transition circuit. The chart construction avoids enumeration of the $N = 2^n$ states. The explicit arithmetic and selection constructions above then compile an m-bit iterator in time polynomial in $n, r+1, m$ with the gate bound of Theorem 7.2.

The executable `sparse_counter.py` implements this chart and interval compiler. Independent finite checks and their scope are recorded in `REVIEW.md`; they do not prove the gate bound or compiler correctness.

7.2.3 Large transitions and charged coordinate changes

The improvement can apply to transitions with genuinely exponential minimum circuit size. Fix r nonempty counter arcs and reconnect them in any single cyclic order. This gives $(r-1)!$ distinct permutations, each differing from the counter at at most r states and having one orbit of length N. For $r = 2^{\lfloor \alpha n \rfloor}$ with fixed $0 < \alpha < 1$, the logarithm of this family size is $\Theta(r \log r)$. The circuit-description bound used in Section 4 implies that some members need $\Omega(r)$ gates, while all have $O(n(r+1))$-gate circuits. Consequently there exist families with

$$\mathrm{C}(F_n) = 2^{\alpha n + O(\log n)}, \qquad \sup_{T \in \mathbb{Z}} \mathrm{C}(F_n^T) = 2^{\alpha n + O(\log n)}.$$

The lower bound is existential counting, not an explicit hard transition. Using only their transition sizes in Theorem 1.1 would instead give an upper bound with exponential scale $2^{(1+\alpha)n/2}$.

Corollary 7.6 (Charged orbit coordinates)**.** *Suppose a bijection $\phi : \{0,1\}^n \to \mathbb{Z}/N\mathbb{Z}$ satisfies $\phi(G(x)) = \phi(x) + 1 \pmod N$. If a permutation F differs from G at r states, then*

$$\mathrm{C}(F^T) \le \mathrm{C}(\phi) + \mathrm{C}(\phi^{-1}) + O(n(r+1)) \quad (T \in \mathbb{Z}).$$

For an m-bit input time, the last term is $O(n(r+m+1))$.

Proof. Conjugate F by ϕ. The exceptional-set cardinality is unchanged, and the reference permutation becomes the counter. Apply Theorem 7.2 and compose with the two coordinate circuits. □

Both coordinate costs are essential. A long orbit does not itself supply cheaply computable coordinates. Conversely, these examples can visit the entire state space: the compressed object is the arrangement of arcs, without a reduction in the number of reachable states.

For the pair queries of Corollary 7.4, coordinate transport needs only the forward map: encode both x and y, then query the conjugated permutation. Reachability and visit times are preserved by conjugacy, so the bound is $2\mathrm{C}(\phi) + O(n(r+1))$. No state is returned, and no inverse-coordinate circuit is required.

A finite-field application, including the cost of discrete-logarithm coordinates, is given in Section B.

8 Lean formalization

The accompanying `lean/` project proves existence of actual Boolean circuits with evaluation semantics and gate bounds. It covers Table 1, the invariant-set, image-sensitive and active-mass refinements, permutation inversion, complete-configuration simulation, and the lower-bound results. The semantic lemmas also support arbitrary state types with finite invariant subsets and supplied Boolean encodings; initialization and decoding costs remain additional.

The development uses CSLib's circuits and simultaneous synthesis relation, Mathlib's dynamics and arithmetic, and the author's Circuit library through `complexitylib`, including its sorting and counting results [11, 12, 13]. Its native basis consists of Boolean constants, NOT, AND, and OR. Every gate is charged, including constants; wiring and fan-out are free. This differs from the paper's basis by an absolute factor and an additive constant. Section A states the native tradeoff and explains proof differences. The public theorem map, detailed constants, module guide, pinned revisions, and build commands are maintained in `lean/README.md`.

The exported theorems have guarded axiom lists containing only `propext`, `Classical.choice`, and `Quot.sound`, with no `sorry` placeholders or additional axioms. Classical choices supply checkpoints, enumerations, and circuit witnesses; the result is nonuniform existence. The sparse chart's existence and circuit cost are checked, but the edit-list compiler's algorithm and runtime, sparse-transition counting application, and finite-field corollary remain prose results. The cryptographic and literature discussions are also outside the mechanized boundary. Kernel checking establishes neither historical priority nor independent fidelity to every intended model convention.

9 Conclusion and open questions

Sparse checkpoints and charged lookup give square-root state-space bounds for arbitrary fixed iterates and functional-graph navigation. The counting example makes the exponent sharp for unrestricted fixed powers; it does not establish sharpness for polynomial-width input times.

Two directions remain: the additive bound $s + \widetilde{O}(2^{n/2})$, and broader classes of transitions with cheaply constructible orbit coordinates. The sparse-counter compiler shows how an explicit structural description can supply such coordinates; finding them from a succinct transition circuit is a separate problem.

Provenance and draft status

This draft and the underlying argument were developed with assistance from ChatGPT (GPT-6 Astra Pro), including assistance with theorem formulation, proof development, exposition, and literature searches. The Lean development and the current revision also received assistance from Codex, including circuit synthesis proofs, axiom auditing, and review. The author field is provisional. Historical priority and final attribution require author review; the draft does not represent independent peer verification.

A Correspondence with the formalization

The native circuit model charges constants as well as Boolean operations. Its foundational statement is the following.

Theorem A.1 (Mechanized tradeoff)**.** *Let c be a circuit over the Boolean constant, NOT, AND, and OR basis, with n inputs and n outputs, and let $S \subseteq \{0,1\}^n$ be finite and invariant under its evaluation F. For all natural numbers T and $1 \le L \le 2^n$, Lean proves the existence of a circuit d agreeing with F^T on S such that*

$$|d| \le 40000 \left[L|c| + (n+1)^3 \left(L + \lfloor |S|/L \rfloor + 1 \right) \right].$$

Here circuit size counts every gate, including constant gates.

The harmless +1s cover zero-width and empty-set cases. Define

$$\widehat{A} = s + (n+1)^3, \qquad \widehat{B} = \widehat{A} + \sqrt{(n+1)^3 2^n \widehat{A}}.$$

The checked real optimization gives the fixed-power bound $120000\widehat{B}$, and a separate theorem derives the $2^{\lceil n/2 \rceil}\,\mathrm{poly}(n)$ consequence for polynomial-size updates. The constants are deliberately loose accounting bounds. Free wiring and arbitrary fan-out permit sharing; constants are charged. Translating the paper's fixed fan-in-two basis to this basis changes size by at most an absolute factor and an additive constant.

The public theorem map in `lean/README.md` gives the remaining exact constants and exported names. Three implementation differences explain how the formalized constructions realize the paper's asymptotic bounds.

1. For a single fixed power, one useful collision suffices. The circuit sorts checkpoint and prefix records with tagged padding, then selects an adjacent equal-key pair witnessing either a checkpoint hit or a repeated state. Batches use the full shared dictionary lookup of Lemma 2.1. The repeated-state decoder adds the period to a hardwired residue until the index lies in the repeated interval; its $O(Kn^2)$ cost, with $K = 2L$, fits the main budget.

2. Cycle labels are computed after projecting by $F^{N!}$, which reaches the cycle and preserves its labels. A short cyclic prefix either meets a checkpoint or contains the whole cycle, allowing a scan for its minimum and period. This replaces the direct first-repetition argument in Theorem 6.1. The forest proof obtains ancestry intervals by ordering root-to-vertex paths, without implementing a traversal algorithm.

3. Sparse-counter charts use at most $r + 1$ ordinary linear intervals, with hardwired canonical cycle labels, rather than compiling the weighted arc permutation. Encoding and decoding handle phase wraparound within those intervals. Both coordinate circuits and one streaming remainder computation are charged. This proves the claimed circuit existence without verifying the edit-list compiler.

Internal arithmetic uses spare bits to represent full-length cycles and prevent overflow. The checked first-hitting-time circuit returns $n + 1$ time bits; its bound below 2^n on positive instances permits discarding the top bit. Visit descriptors use a success flag and two $(n + 1)$-bit fields, with period zero denoting a singleton. Their all-time semantics and bounded-count interpretation do not include an additional deadline-processing circuit.

B Sparse edits of a primitive binary-field cycle

Corollary B.1. *Represent $K = \mathbb{F}_{2^n}$ in a fixed polynomial basis and fix a primitive element ω. If a permutation F differs from $L(x) = \omega x$ at r states, then*

$$\mathrm{C}(F^T) \le n^{O(\log n)} + O(n(r+1)) \quad (T \in \mathbb{Z}).$$

With an m-bit input time the bound is $n^{O(\log n)} + O(n(r+m+1))$.

Proof. Insert zero into the nonzero multiplicative cycle by setting $G(0) = 1$, $G(\omega^{-1}) = 0$, and $G(x) = \omega x$ elsewhere. Then G is one cycle, and F differs from it at at most $r+2$ states. Coordinates are $\phi(0) = 0$ and $\phi(\omega^j) = j + 1$ for $0 \le j < 2^n - 1$. Decoding uses finite-field exponentiation and costs $\mathrm{poly}(n)$ gates. Encoding uses discrete logarithms.

Kleinjung and Wesolowski's Theorem 1.1 [10] gives expected time $n^{O(\log n)}$ for discrete logarithms in characteristic two. To obtain exact nonuniform circuits, let $\tau(n)$ bound this expectation for every input. Truncate each trial at $2\tau(n)$ steps and verify candidate logarithms by exponentiation. A trial succeeds with probability at least $1/2$. Among $n+1$ independent random tapes, the probability that some one of the fewer than 2^n nonzero inputs has no successful trial is less than one. Fix tapes that work for every input and select the first verified answer. Circuit simulation and verification have polynomial overhead in $\tau(n)$ and n, preserving quasipolynomial size. Apply Corollary 7.6. The case of bounded n is absorbed by the constants. □

This corollary relies on the cited discrete-logarithm theorem and the nonuniform simulation argument; those ingredients are not formalized in our Lean development. It concerns sparse permutation-preserving edits to a primitive multiplicative cycle. It does not imply the unrestricted conjectural bound $s + \widetilde{O}(2^{n/2})$.

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